Tshana Tumelo Konaite
Software Engineer
Email: **tumelokonaite@gmail.com** | **GitHub** | **Portfolio** | **LinkedIn**

Personal Statement

Software Engineer with a strong academic background in Computational and Applied Mathematics and hands-on experience designing, developing, testing, and deploying production software. Experienced in Python, FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Docker, CI/CD, cloud platforms, automated testing, and Infrastructure as Code. Comfortable building features end to end, debugging applications, integrating external systems, and maintaining reliable and scalable backend services. Also brings experience in AI and machine learning, providing a strong analytical approach to problem-solving and software development.

CORE SKILLS & TECHNOLOGY STACK

MLOps and Model Operations: MLflow, model deployment

Cloud, DevOps and Infrastructure: AWS, GCP, Huawei Cloud, Docker, GitHub Actions, Terraform, CI/CD

Machine Learning and Data Processing: Scikit-learn, Pandas, NumPy

AI Engineering and Generative AI: Generative AI, LLM integration, RAG, LangGraph, LangChain, prompt

Backend and API Engineering: FastAPI, SQLAlchemy, Pydantic, REST APIs, OpenAPI, webhooks, Kafka

Monitoring and Observability: Arize AX, Evidently, MLflow, production logging, prediction tracking, delayed-

LLM Platforms and Vector Search: OpenAI API

Testing and Tools: Pytest, Unittest, Linux, Git, Postman, Swagger, Bash scripting

Databases and Caching: PostgreSQL, MySQL, Redis

Programming Languages: Python, SQL, JavaScript, HTML5, CSS3, Bash

PROFESSIONAL EXPERIENCE

Software Engineer Softwareverse Dec 2025 – Present

Core Responsibilities:

- Design, develop, and deploy scalable backend applications and AI-powered software solutions using Python, FastAPI, PostgreSQL, and REST APIs.
- Design and implement cloud-native application architectures, integrating authentication, databases, external APIs, and supporting infrastructure.
- Implement MLOps workflows for model experimentation, evaluation, deployment, prediction tracking, and performance monitoring using MLflow and related tools.
- Containerise applications using Docker and automate testing, infrastructure provisioning, and deployment through GitHub Actions, Terraform, and cloud services.
- Develop automated unit and integration tests, troubleshoot production issues, and maintain reliable, secure, and scalable software systems.

Transmission Engineer Huawei **Technologies Dec 2022 –Dec 2025**

Core Responsibilities:

- Led technical delivery across multiple telecommunications projects, working closely with customers to
- Led technical delivery across multiple telecommunications projects, working directly with customers to understand business requirements, define technical solutions, and coordinate engineering teams.

- Designed, developed, and deployed an AI-powered internal knowledge assistant using departmental documentation and operational data to support employee onboarding and technical knowledge retrieval.
- Deployed and maintained the AI assistant and supporting backend services on Huawei Cloud, supporting enterprise knowledge management and internal operations.
- Participated in technical solution design, proof-of-concept development, feasibility assessments, and customer presentations to support technical decision-making.

Data Analyst Huawei Technologies Dec 2019 –Dec 2022

Core Responsibilities:

- Collected, processed, and analysed operational and customer datasets to identify performance trends, customer behaviour patterns, and opportunities for process improvement.
- Explored predictive machine learning models to identify customers likely to make repeated service contacts, applying classification techniques to support proactive customer service strategies.
- Conducted exploratory data analysis, data preprocessing, feature engineering, and model evaluation to support analytical investigations.
- Developed performance reports and data-driven insights to support operational planning, resource allocation, and management decision-making..

PROFESSIONAL CERTIFICATIONS

2026 – AWS Certified Generative AI Developer – Professional (Early Adopter)

2026 – AWS Certified Solutions Architect – Associate

2026 – SAS: Certified Associate – Programming Fundamentals Using SAS 9.4

2025 – AWS Certified Machine Learning Engineer – Associate

2025 – AWS Certified AI Practitioner

EDUCATION

University of the Witwatersrand | Feb 2021 – Nov 2024

MSc in Computational and Applied Mathematics

University of the Witwatersrand | Feb 2019 – Nov 2019

BSc Honours in Mathematical Sciences